

T.POOJITHA

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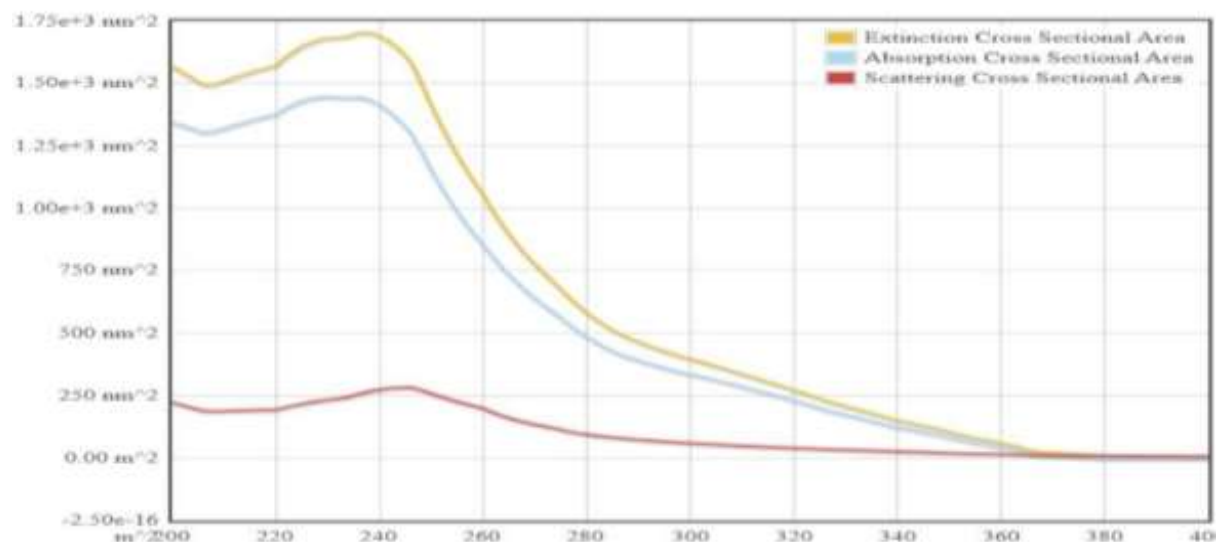
ECA3102 NANO ELECTRONICS

Exp. 1. Simulation of scattering and absorption of Multi-layer Sphere and Core-Shell Nanoparticle using Mie-Plot tool

Aim:

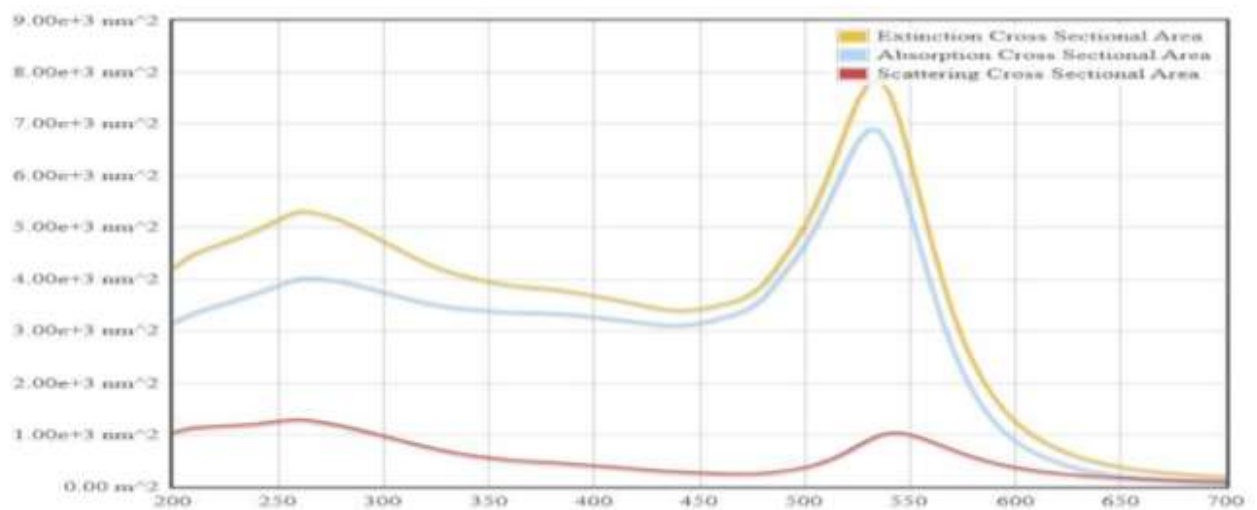
To simulate and analyze the scattering, absorption, and extinction characteristics of multi-layer spheres and core-shell nanoparticles using the Mie-Plot tool.

Case Study 1: Scattering and absorption spectra for a single-layer sphere (core only).



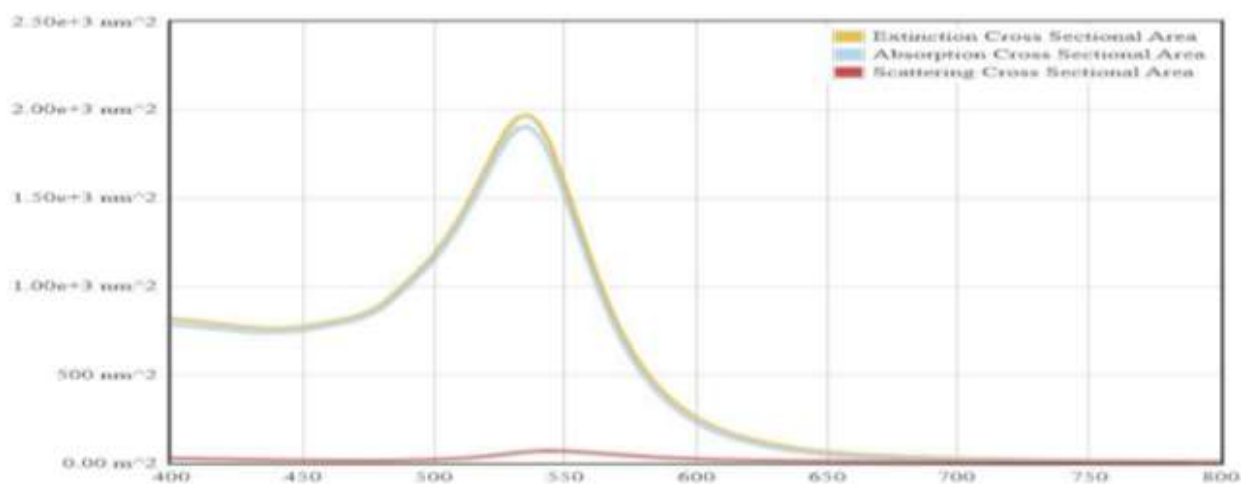
The graph shows the extinction, absorption, and scattering cross-sectional areas of a 50 nm gold nanoparticle in water as a function of wavelength, with maximum optical response occurring near 240 nm.

Case Study 2: Simulate the optical response of a gold (Au) nanoparticle of 50 nm diameter in water medium.



The graph shows the extinction, absorption, and scattering cross-sectional areas of a gold-core (30 nm) silica-shell (10 nm) nanoparticle, with the highest optical response occurring around 540–550 nm.

Case Study 3: Simulate a core-shell nanoparticle with a gold core (30 nm) and silica shell (10 nm).



The graph shows the extinction, absorption, and scattering cross-sectional areas of a 25 nm gold nanoparticle, with the maximum optical response occurring around 540 nm.